

Exercise 8: Performance and Publishing

Three worksheets, three captions. No separate report.

Item	Details
Points	100
Data	clean/mvu_student_dim.csv and clean/mvu_enrollment_fact.csv
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 2 (extracts) and Ch 13 (data model). Tableau Help: Record and Analyze Workbook Performance; Extract Your Data.
Walkthrough	Module 8 of the MVU Tableau Walkthrough follows these same steps, with checkpoints.
Submit	One packaged workbook (.twbx) with Worksheets 1–3 and a caption on each, plus Performance_Before.twbx and Performance_After.twbx.

Before you start

The MVU enrollment table has about 10,000 rows, so most views will redraw in well under a second. That's expected. The skill is reading the evidence: which event is slowest, and which design choice caused it. The same causes that cost milliseconds here cost minutes on fifty million rows.

Checkpoint: How many rows are in the enrollment fact table?

Worksheet 1: diagnose a slow view

1. Join mvu_student_dim and mvu_enrollment_fact with an inner join on student_id (physical layer, not a relationship).
2. Create Course Label = UPPER(TRIM([course_code])) + " — " + UPPER([course_title]) + " (" + [term_code] + ")".
3. Sheet Worksheet 1 — Before: student_id and term_code on Rows; SUM(credit_hours) on Text; Course Label on Detail.
4. Add full_name as a quick filter set to Only Relevant Values. Note: 933 names for 1,400 students, so names are shared.
5. Put the sheet on a dashboard with a headcount-by-college sheet and a filter action from it.

Checkpoint: How many marks does the Before sheet draw (status bar, bottom left)?

6. Help > Settings and Performance > Start Performance Recording. Click three colleges, change the name filter twice, switch sheets and back. Stop the recording and save it as Performance_Before.twbx.
7. In the Performance Summary, read the Events view (longest first) and the Query view for the slowest Executing Query event.
8. Fix one thing at a time: a relationship instead of the join; student_id (or no filter) instead of the name filter; remove Course Label from Detail; ask whether a summary by college and term would serve the reader.
9. Save the result as Worksheet 1 — After, record the same interactions again, and save Performance_After.twbx.

Worksheet 2: extract only what the view needs

1. New data source: mvu_enrollment_fact.csv alone. Worksheet 2: college on Rows, term_code on Columns, a record count on Text.
2. Switch Live to Extract and click Edit. Filter cohort_year to 2020 and 2021. Click Hide All Unused Fields, then tick Aggregate data for visible dimensions.
3. Create the extract, then Data > your data source > Extract > Compute Calculations Now.
4. Decide which column an incremental refresh would use, and where incremental refresh would be wrong.

Checkpoint: How many enrollment records remain after the cohort filter?

Worksheet 3: publish and govern

1. Hide unused fields, check container padding, name every sheet. Build a small dashboard, Worksheet 3, from Worksheet 1 — After and Worksheet 2.
2. Server > Publish Workbook to the course project your instructor provides. Viewers can view; only you can edit. Include the extract with the workbook.
3. Open the published view in the browser and check it matches Desktop.

No server access? Write the settings you would choose, and why, in the caption. The reasoning is what's graded.

Your three captions

Sheet	What your caption must answer
Worksheet 1	What was the slowest event before, and which design choice caused it? What did you change, and what did the second recording show? What would the same design cost on fifty million rows?
Worksheet 2	What did you filter, hide and aggregate, and how many records remain? Where would incremental refresh be safe in MVU's data, and where would it be wrong?
Worksheet 3	Who can see and who can edit the published view, and why? How will its data stay current, and does embedding credentials apply here?

Worksheet 1's caption carries 25% on its own. "The After version is faster" does not earn full marks. Name the event, name the cause, and show the change one fix made.

Before you submit: confirm every caption is visible.

Submit for Exercise 8

- Worksheet 1: Before and After sheets, with Performance_Before.twbx and Performance_After.twbx
- Worksheet 2: a filtered, aggregated extract with unused fields hidden
- Worksheet 3: the published view's link, or your chosen settings if you couldn't publish
- A visible caption on each
- Packaged workbook (.twbx). No separate report.